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Affiliation errors and distorted researcher mobility: evidence from OpenAlex and Scopus

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Abstract

Bibliographic metadata underpin large-scale studies of researcher mobility by enabling affiliation timelines to be reconstructed from publication records. Yet the field lacks a systematic benchmark of false-positive affiliations and how they distort mobility indicators, especially at institutional granularity. We evaluate OpenAlex and Scopus using a cohort from the 2020 Neural Information Processing Systems conference. Our two-stage audit proceeds as follows. First, we construct institution- and country-level timelines for a sample of 100 authors using a mode-per-year rule and validate yearly states and reported moves. Second, we manually review 2093 publications for a 25-author sample, verifying each author–publication–affiliation link and classifying errors as author disambiguation or affiliation extraction. Three findings stand out. Institution-level timelines are less accurate than country-level timelines in both platforms: average author-weighted affiliation accuracy is about 0.62 versus 0.75 in OpenAlex and about 0.70 versus 0.86 in Scopus. Platform aggregates favor Scopus, but within-author comparisons on overlapping years show much smaller differences (about −0.03 at institutions and −0.07 at countries, with intervals including zero), indicating composition and coverage effects. Many reported migration events lack two verified endpoints: positive predictive value is about 0.29 in OpenAlex and 0.45 in Scopus at institutions, so raw counts overstate mobility. Error mechanisms differ by platform: in OpenAlex, errors arise from both identity and affiliation extraction; in Scopus, affiliation extraction dominates (about 1.25 errors per publication versus 0.06 in OpenAlex). We provide a cross-platform benchmark and a replicable protocol, and recommend reporting event validity, using simple concordance checks, and prioritizing mechanism-targeted improvements.

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Introduction

Understanding the mobility of researchers across institutions, sectors, and countries is central to the study of contemporary science systems (Robinson-García et al., 2019; Wagner et al., 2018). Patterns of movement not only reflect individual career trajectories but also capture the broader circulation of expertise, the redistribution of human capital, and the institutionalization of collaborations across borders (Edler et al., 2011; Jonkers & Tijssen, 2008; Momeni et al., 2022). As such, the study of mobility illuminates how knowledge flows reinforce or challenge national capacities in strategically important areas such as artificial intelligence (AI), biotechnology, or climate science (Chinchilla-Rodríguez et al., 2018; Huang et al., 2024)—in this study, we focus on researchers in AI, a field that is both highly internationalized and policy-relevant. From the perspective of science policy, mobility metrics shape debates on talent retention, international competitiveness, and the allocation of research funding (Fontes, 2007). Bibliometric methods have become the dominant approach for studying these issues at scale (Akbaritabar et al., 2024; Moed & Halevi, 2014). By reconstructing affiliation timelines from publication metadata, researchers infer institutional and geographic transitions across entire populations of scholars (Robinson-García et al., 2019). This approach has yielded influential findings, such as patterns of brain drain from developing economies, the emergence of transnational research communities, and the role of mobility in knowledge transfer (Jonkers & Tijssen, 2008; Meyer, 2001).

Large-scale analyses of researcher mobility rely on bibliographic platforms that aggregate publication metadata and link works to authors and institutions. Among these, OpenAlex and Scopus are two of the most widely used. OpenAlex provides fully open and reproducible access to works, authors, venues, and institutions, making it a valuable resource for transparent bibliometric research (Alperin et al., 2024; Priem et al., 2022). Scopus, by contrast, is proprietary but long-established, employs structured metadata such as rich institutional hierarchies (Elsevier, 2023; Griffiths, 2019), and offers broad coverage of journals and conference proceedings, with particular strength in computer science and AI (Mongeon & Paul-Hus, 2016; Singh et al., 2021; Zhang, 2014). Taken together, OpenAlex and Scopus represent complementary perspectives: one prioritizing openness and reproducibility, the other emphasizing structured metadata and disciplinary coverage.

Even the most reliable bibliographic platforms exhibit weaknesses that threaten the accuracy of inferred mobility timelines. A central issue is author name disambiguation—algorithms rely on imperfect name-matching heuristics, leading to both over- and under-clustering of identities, especially in fields and regions with common names (D'Angelo & van Eck, 2020; Kim & Owen-Smith, 2021; Torvik & Smalheiser, 2009; Zhou & Sun, 2024). Similarly, affiliation metadata often suffer from parsing errors, inconsistent naming conventions, and missing institutional identifiers, complicating cross-platform harmonization (Lin et al., 2024; Purnell, 2022; Zhang et al., 2024). Although these platforms achieve broad coverage and high recall (Culbert et al., 2025; Singh et al., 2021), the trade-off is reduced precision, as noisier or inconsistently curated records can produce false-positive affiliations (Zhao & Chen, 2025). These challenges are not unique to a single platform but intrinsic to the design of bibliographic systems, underscoring the need for validation strategies that address identity, affiliation, and precision errors in concert (Donner, 2024).

Errors at the level of author–publication–affiliation metadata propagate mechanically into mobility timelines: when authorship is misassigned or an affiliation string is incorrect, the resulting series records a move that never occurred or misses one that did (Donner, 2024; Robinson-García et al., 2019). Aggregation to coarser units can mitigate some of this noise—country-level, year-level reconstructions often absorb within-country variation—whereas fine-grained, institution-level timelines (and subnational geographies) are far more sensitive to spurious labels and within-year misplacements (Akbaritabar et al., 2025; Donner et al., 2020). Because false positives add affiliations rather than merely omitting them, they create asymmetric bias: they fabricate transitions and thus risk systematically inflating mobility indicators.

Despite pervasive metadata imperfections, the field still lacks a systematic benchmark for the prevalence and mechanisms of false-positive affiliations. We address this by quantifying false-positive rates across two major bibliographic platforms and identifying the processes that generate them. Using an AI cohort from NeurIPS 2020 (Larochelle et al., 2020), we test two hypotheses: (H1) false-positive affiliations systematically inflate mobility; and (H2) this inflation is larger at the institution level than at the country level. We chose NeurIPS because it is one of the largest and most internationally prominent venues in machine learning and AI, attracting authors from major research institutions worldwide across both academia and industry. AI researcher mobility is itself a highly active and policy-relevant topic of study, adding substantive motivation to the methodological focus. We selected the 2020 proceedings because they provide a complete and consistently published set of papers with finalized author affiliations.

We implement a two-part validation framework. First, we audit institution- and country-level timelines for a random sample of 100 NeurIPS 2020 authors using a mode-per-year rule (Subbotin & Aref, 2021; Zhao et al., 2022). Second, for a 25-author sample, we perform a publication-level review that verifies author–publication–affiliation links and attributes errors to disambiguation or affiliation-extraction mechanisms. This design yields four contributions: (1) empirical estimates of inflation in researcher mobility (H1), with effect sizes at the institution and country level; (2) evidence for stronger inflation at the institution level than the country level (H2); (3) a mechanism-resolved benchmark of error rates in author–publication–affiliation links; and (4) a replicable validation protocol that integrates large-scale metadata with targeted manual verification and uncertainty quantification.

Literature review

Researcher mobility has been investigated extensively through bibliometric data, but the literature spans diverse traditions that emphasize different levels of analysis and methodological concerns. To situate our contribution, we organize this review around four strands most relevant to the present study: (i) bibliometric foundations for inferring mobility from publication affiliations; (ii) metadata quality issues, including author disambiguation and affiliation parsing; (iii) coverage and selectivity differences across major bibliographic platforms; and (iv) validation, uncertainty, and the role of persistent identifiers. Together, these strands provide the empirical and methodological context for our two-part audit of false-positive affiliations.

Bibliometric foundations: inferring mobility from publication affiliations

Bibliometric studies infer mobility by tracking changes in authors' affiliation addresses on publications. Early large-scale implementations in Scopus, for example, were developed by Moed and Halevi (2014), who outlined both synchronous and asynchronous migration indicators and demonstrated the feasibility—and limits—of this approach. Building on these foundations, Robinson-García et al. (2019) consolidated conceptual categories of mobile researchers (e.g., "migrants" versus "travelers") and established standard practices for turning author–affiliation linkages into mobility types and flows.

A now-dominant operational choice is to assign each author a mode country (or region) of affiliation per year and to log a migration event when the mode changes between years; tie-breaking typically prefers the prior mode to stabilize series. This rule has been applied at both global and national scales. Akbaritabar et al. (2024, 2025) formalized its use for international and subnational flows, while Zhao et al. (2022) applied it to German researchers, registering a migration event only if the previous mode disappears in the new year. Country-level and subnational implementations further adopt pragmatic assumptions to handle publication delay and gaps (e.g., 2-year padding and last-observation carry-forward), which reduce volatility at yearly resolution (Akbaritabar et al., 2025).

These design choices support both macro- and meso-level indicators. Event counts can be aggregated into bilateral flows and rate measures such as the net migration rate (NMR), facilitating comparisons across regions and years (Akbaritabar et al., 2025). National applications illustrate the framework's empirical reach—for instance, Subbotin and Aref (2021) analyzed Russia's researcher migration dynamics (including robustness checks on padding and data exclusion), while Zhao et al. (2022) examined Germany's gender-differentiated mobility patterns.

In sum, the literature has converged on mode-based, year-level reconstructions of affiliation geography—applied from global (Akbaritabar et al., 2024) to subnational scales (Akbaritabar et al., 2025)—providing a consistent baseline against which to assess how metadata errors propagate into mobility inferences.

Metadata quality, disambiguation, and affiliation parsing

Inferring mobility from publication data depends on the correctness of three linked elements at the record level: the author identity on a work, the affiliation(s) attached to that authorship, and the ability to harmonize those fields across platforms. Errors in any of these elements propagate directly into inferred transitions, particularly when timelines are constructed at fine (institution) resolution.

A long line of work addresses *author name disambiguation* (AND), highlighting both over- and under-clustering risks in large bibliographic corpora. In biomedical metadata, Torvik and Smalheiser (2009) developed a probabilistic "Author-ity" model that integrates article-level features (coauthors, title terms, affiliations, MeSH, etc.) to cluster namesakes with high recall, while documenting residual lumping and splitting error modes. At the scale of field-wide or career-level retrieval, D'Angelo and van Eck (2020) propose practical AND procedures for assembling per-researcher publication sets and discuss trade-offs that arise from heuristic thresholds. For mobility studies, the key implication is that over-merged profiles concatenate distinct careers and fabricate moves; under-merged profiles break real careers and obscure moves.

Beyond identity, *affiliation extraction and normalization* present distinct challenges. Comparing four major databases, Purnell (2022) quantifies substantial cross-platform disagreement in university assignments for the same DOI sets and attribute discrepancies to missing affiliation strings, divergent unification rules, and misassignments; in one matched comparison he identifies 30,060 records present in Scopus but not in WoS, 1831 records in WoS with a *different* university assignment, 1661 with missing affiliations, and 1094 with sufficient metadata to reconcile disagreements. Recent NLP approaches combine named-entity recognition with semantic and geographic signals to improve mapping of free-text affiliation strings to persistent organization identifiers (e.g., ROR), reducing synonym/abbreviation and location ambiguities (Lin et al., 2024). Platform-specific assessments also matter. In an OpenAlex–Scopus comparison, Alperin et al. (2024) report that while OpenAlex often matches or exceeds Scopus' coverage, several metadata fields (including institutions) exhibit completeness and correctness issues. Complementing this, Zhang et al. (2024) document non-trivial rates of missing institutional information in OpenAlex journal-article records, with systematic variation by field and publication year, and discuss implications for university- and country-level analyses.

These strands motivate our validation design: we explicitly separate (i) identity errors (misassigned works) from (ii) affiliation parsing/mapping errors on correctly attributed works, and we quantify how each mechanism affects institution- and country-level timelines.

Coverage and comparative biases in major bibliographic databases

Large-scale mobility studies typically draw on multidisciplinary indexes with curated journal lists. Among these, Web of Science (WoS) and Scopus have been the dominant platforms, but they differ in selectivity and scope. Comparative analyses show that Scopus indexes a substantially larger set of active journals than WoS, with systematic differences by field, language, and publisher country. In particular, Natural Sciences & Engineering and Biomedical Research are overrepresented in both databases relative to Ulrich's, while Social Sciences and Arts & Humanities are underrepresented; English-language journals and publishers from the United States, United Kingdom, and the Netherlands are comparatively overrepresented (Mongeon & Paul-Hus, 2016).

At the journal level, most WoS titles are contained in Scopus: Singh et al. (2021) report that *99.11%* of WoS journals were also indexed in Scopus in their comparison, while Scopus additionally includes many exclusive titles. These structural differences matter for mobility inference: database choice can shift the observable footprint of disciplines and regions (e.g., Scopus's broader inclusion of regional outlets), with downstream consequences for measuring institutional and cross-national flows, especially in fields that rely heavily on conference proceedings such as computer science (Zhang, 2014).

Validation and uncertainty in bibliometric indicators

Even when mobility is operationalized consistently, indicator validity depends on the accuracy of the underlying author–publication–affiliation metadata and on how errors propagate to higher levels of aggregation. A growing line of work stresses the need to quantify data inaccuracies and to assess their impact on derived indicators through explicit uncertainty analyses. Donner (2024) formalizes procedures for diagnosing data errors and propagating them to indicator uncertainty, recommending transparency about sensitivity to input choices and error sources. At the institutional scale, Donner et al. (2020) show that alternative affiliation disambiguation systems produce non-trivial differences in institutional performance indicators, underscoring how affiliation normalization choices translate into measurable output variation. Cross-database studies likewise document disagreement in affiliation assignments for the same works, attributing discrepancies to missing strings, divergent unification rules, and misassignments (Purnell, 2022). Recent evaluations of author identity systems complement these findings by examining precision/recall trade-offs and where misassignments tend to arise: Zhao and Chen (2025) report precision degradation in name-dense contexts and variation by name region, while Zhou and Sun (2024) flag likely disambiguation failures via anomaly analysis of career sequences.

Persistent identifiers and complementary infrastructures

External identifier systems and curated records can strengthen identity and affiliation inference when linked to bibliographic metadata. Kim and Owen-Smith (2021) introduce an ORCID-linked labeled dataset for evaluating author name disambiguation at scale, highlighting the value of researcher-verified identifiers for precision benchmarking. Adoption itself is heterogeneous across disciplines and institutions, with evidence that local policies and journal requirements materially influence uptake (Porter et al., 2025). On the affiliation side, mapping free-text strings to persistent organization identifiers can reduce synonym and abbreviation ambiguities; an end-to-end approach that combines named-entity recognition with disambiguation to registry IDs (e.g., ROR) demonstrates substantial gains in extracting and normalizing affiliations from publications (Lin et al., 2024). In practice, these infrastructures are highly useful for validation and harmonization, but their partial and uneven coverage means they cannot fully substitute for systematic audits when constructing mobility timelines at scale.

Synthesis and unresolved gap

In summary, prior work has converged on mode-based affiliation timelines and has catalogued major sources of error—identity, affiliation parsing, and cross-database discrepancies—as well as structural differences in coverage across platforms. Validation studies show that such imperfections can materially alter institutional and national indicators (Donner, 2024; Donner et al., 2020; Purnell, 2022). What remains missing, however, is a systematic, mechanism-resolved assessment of *false-positive* affiliations and how they inflate mobility metrics, particularly at the institution level. Establishing such a benchmark is essential for gauging the reliability of bibliometric mobility studies and for designing corrective strategies.

Methods

To address this gap, we designed a two-part manual audit of OpenAlex and Scopus. First, we applied a mode-per-year rule (Subbotin & Aref, 2021; Zhao et al., 2022) to construct institution- and country-level timelines for a random sample of 100 NeurIPS 2020 authors and assessed their year-by-year accuracy. Second, we reviewed 2,093 publications for a 25-author sample, examining every author–publication–affiliation instance and categorizing incorrect records as either disambiguation errors or affiliation-extraction errors.

Together, these procedures provide both prevalence estimates of false positives in mobility timelines and insight into the metadata mechanisms that generate them.

Data collection

We began by programmatically harvesting the NeurIPS 2020 proceedings (Larochelle et al., 2020), yielding 1898 unique paper titles and 5916 unique author names, with 1150 authors appearing on multiple papers (maximum of 12). Each author was then linked to OpenAlex and Scopus profiles in order to retrieve their publications and reported affiliations. Because OpenAlex does not index the full proceedings, we first queried its API by year (2020) and the NeurIPS source identifier, returning 1157 candidate records. Because our framework targets false-positive affiliations—incorrect records present in a platform—rather than false-negative coverage gaps, each platform is evaluated only on the records it indexes. Coverage shortfalls, such as OpenAlex indexing 1157 of the 1898 proceedings papers, represent a distinct dimension of data quality that requires different detection strategies and is not reflected in the error rates reported here. Titles from the proceedings and from platform APIs were normalized (Unicode normalization, lowercasing, punctuation removal, whitespace compression) and compared with a character-based similarity score in [0, 1]; for each proceedings title, the best match above a threshold of 0.60 was accepted. Effectively all paper titles were matched with a perfect score of one. Remaining unmatched titles were queried individually by title against OpenAlex and accepted if the top result exceeded the same threshold. Scopus indexes the complete proceedings; we therefore queried its API by year and source identifier and applied the same normalization and threshold to match titles. Platform provenance was retained for all matched records.

To link authors to platform profiles, we normalized names in the proceedings and platform records to last name plus first initial. Within each matched paper, we selected the platform author string with the highest similarity to the normalized proceedings name and accepted the link if the ratio exceeded 0.60, drawing only from the candidate pool of authors listed on that matched paper (typically one to five names). To assess the choice of threshold, we re-ran this matching step post hoc for the Part 1 sample and examined the distribution of similarity scores. The accepted author–profile links were strongly bimodal: 92% scored 0.90 or above, none scored between 0.80 and 0.90, and every rejected best match scored 0.60 or below, so moving the cutoff anywhere between 0.60 and 0.90 changes the status of at most five links. Manual inspection of those five borderline links found three to be spurious, in each case because the focal author was absent from the platform's parsed co-author list for the paper, leaving the closest remaining name to be accepted. Such borderline links are therefore unreliable; they are few, and because a spurious link makes the affected author's timeline read as incorrect, their effect is to modestly inflate rather than understate the OpenAlex error rates reported below. The resulting profile identifiers were then used to retrieve each author's publication list from the respective platform.

For every retrieved publication, we collected the affiliations reported for the focal author. In OpenAlex, affiliations mapped to known institutions were stored by the platform's display name; affiliations not mapped to known institutions were stored as raw affiliation strings. In Scopus, affiliations are returned with structured fields—institution name, city, and country—within a hierarchical parent–child structure that can extend to departments or labs; because department-level nodes are not always explicitly linked to their parent institution, we additionally preserved the reported city and country for each Scopus affiliation to support downstream harmonization during manual review. All affiliations retained their platform label. The union of affiliations attached to all publications for a given author constituted the set evaluated in the second part of our analysis.

From the 5916 unique author names in the NeurIPS 2020 proceedings, we drew a simple random sample of 100 authors using a fixed seed for reproducibility for the first part of the audit. A simple random sample of 25 authors was then drawn for the publication-level review; this size was chosen to balance mechanism-level insight against the cost of manually verifying each author–publication–affiliation instance (2093 publications in total).

For the first part of our analysis, we constructed separate annual timelines from OpenAlex and from Scopus for each author–platform pair. We defined an active range as the inclusive interval from the first to the last publication year recorded by that platform; only years within this range were populated. For each in-range year, we aggregated all affiliations on the author's works and applied a mode-per-year rule (Subbotin & Aref, 2021; Zhao et al., 2022) independently at the institution level and at the country level. If multiple values tied for the mode, we carried forward the previous in-range year's value to stabilize the series; if no prior value existed (the first in-range year), ties were broken deterministically by lexical order. Years within the active range that contained no publications inherited the previous in-range year's institution and country (last-observation-carried-forward). Raw, unmatched OpenAlex affiliation strings were excluded from the mode computations because they would require additional parsing beyond the scope of this study. This exclusion may create an asymmetry relative to Scopus, where affiliation records consistently include structured fields (institution name, city, and country). To quantify the scope of this exclusion, we conducted a post hoc re-query of the OpenAlex API. For each Part 1 author whose OpenAlex profile could be re-resolved and verified against the originally audited timeline—68 of the 83 audited OpenAlex authors—we retrieved every indexed publication and classified each as carrying a mapped institution, only a raw (unmapped) affiliation string, or no affiliation data of any kind. Of the 1141 verified OpenAlex author-years, 182 (16.0%) contained at least one publication with only a raw affiliation string, distributed across 40 of the 68 authors; of these, 8 author-years (0.7%), spanning 8 authors, consisted entirely of such publications. Because a raw-only publication contributes no institution to the annual tally, the exclusion alters the computed institution mode only in those 8 fully affected author-years, which fall back to carry-forward; in the remaining partially affected years the mode is still determined by the author's mapped publications. Annual values were thus produced for each platform.

Error definitions and review scope

We evaluated metadata errors at two levels aligned with the study design. At the *timeline* level, for each author–year and platform within the platform's active range, we assessed whether the modal institution and the modal country were correct relative to external evidence. Years outside a platform's active range were ignored by construction.

At the *publication* level (25-author sample), we examined every author–publication–affiliation instance and labeled each as correct or incorrect. Incorrect instances were then attributed to one of two mechanisms that directly generate timeline errors: (i) *disambiguation errors*, in which a publication is misattributed to the focal author, rendering all affiliations on that authorship spurious (Torvik & Smalheiser, 2009); and (ii) *affiliation-extraction errors*, in which authorship is correct but the institutional metadata for that authorship are wrong (e.g., malformed or mismapped institution strings, mis-assigned child units, coauthor spillover)

(Purnell, 2022). This mechanism-resolved taxonomy matches the identity versus affiliation parsing/normalization distinction used in prior work.

To avoid masking platform-specific failure modes, we retained OpenAlex affiliations that lacked institution identifiers (raw strings) and preserved the full Scopus affiliation strings together with their reported city and country during the publication-level review. For cross-platform alignment in descriptive comparisons, we used the DOI where available, but we did not merge or reconcile affiliation strings across systems because representations and granularity differ substantively. Background sources of noise such as missing works in a profile, duplicate records, or name variation are acknowledged but were not directly quantified here, as they require distinct detection strategies (e.g., false-negative publications cannot be confirmed by definition).

Manual audit protocol

For the 100-author timeline study (Part 1) and the 25-author publication-level audit (Part 2), we reconstructed each focal author's institutional history from publicly verifiable sources and compared platform reports to that reconstruction. Evidence was used in the following order of precedence: publication PDFs and publisher landing pages (affiliation as printed on the work), official university or laboratory pages, author curricula vitae (when available), ORCID profiles, and, if needed, secondary sources such as LinkedIn. Because affiliation timelines are annual, appointment intervals were coded at year resolution.

For a given year t, an institution reported by a platform was counted as *correct* if it matched any institution the author listed on their publications in year t or any verified appointment spanning any portion of year t. When publications listed multiple concurrent affiliations, any of those institutions qualified as correct. For Scopus entries naming department- or lab-level units, a match to the documented parent institution was also considered correct; conversely, a parent reported by a platform was considered correct if the publication listed a recognized child unit of that parent. Country-level correctness was defined by the ISO 3166 country of the verified institution.

For each author–publication–affiliation instance in the 25-author sample, labeling proceeded in two steps. First, we verified authorship; if external evidence indicated that the publication did not belong to the focal author (e.g., different identity despite similar name), the instance was labeled a *disambiguation error*. If authorship was confirmed, we compared the platform's affiliation for the focal author against the affiliation(s) listed on the publication for that author; mismatches were labeled *affiliation-extraction errors*. Platform provenance (OpenAlex vs. Scopus) was retained in all labels.

Ambiguous cases were handled conservatively. Instances lacking minimally sufficient evidence to adjudicate (e.g., no affiliation on the publication and no verifiable appointment for the year) were not marked as errors. Authors with no retrievable publications in a platform were retained to reflect real-world coverage gaps but contributed no timeline observations for that platform. This protocol yields, for each platform, author–year correctness indicators at the timeline level and counts of disambiguation and affiliation-extraction errors at the publication level, which form the basis for the statistical analyses that follow.

Statistical estimands and uncertainty quantification

This subsection defines the summary measures we report for the timeline audit (Part 1) and the publication-level audit (Part 2). All computations are performed separately at the institution level and the country level, and separately for OpenAlex and Scopus.

Part 1: Timeline audit

For each author and platform, we first compute the share of years in which the platform's modal affiliation is correct. The *author-weighted average accuracy* is the mean of these per-author shares across all eligible authors. The *year-weighted average accuracy* is the corresponding average across all author–year observations (i.e., the fraction of all in-range author–years that are correct). We also report the *proportion of authors with any incorrect affiliation*, defined as the fraction of eligible authors who have at least one incorrect year in their in-range timeline.

To directly compare platforms on the same footing, we compute an *average paired difference* between OpenAlex and Scopus. For each author with at least one overlapping in-range year in both platforms, we take the difference in accuracy between OpenAlex and Scopus computed over those overlapping years, then average that difference across such authors. Positive values indicate higher accuracy for OpenAlex on the overlapping years; negative values indicate the reverse.

To link errors to mobility inflation, we evaluate migration events. For each platform and author, we identify year-to-year mode changes (migration events) within the platform's active range. The *event-level positive predictive value for migration* is the fraction of reported migration events for which both the origin year (t) and the destination year ($t + 1$) affiliations were correct. Its complement approximates the share of reported events that are likely spurious. We compute this separately at the institution and country levels.

Part 2: Publication-level audit

For the 25-author sample, every author–publication–affiliation instance is labeled as correct or incorrect, and incorrect instances are further attributed to disambiguation errors or affiliation-extraction errors. For each platform and for each error grouping (all false positives combined; disambiguation only; affiliation-extraction only) we report three quantities. The *author-weighted error rate* is the mean, across authors with at least one publication, of each author's count of incorrect affiliation instances divided by their number of publications. The *publication-weighted error rate* is the total count of incorrect affiliation instances divided by the total number of publications, pooled across authors. Because a single publication may carry multiple affiliations for the focal author, both rates can exceed one. The *proportion of authors with any incorrect affiliation* is the fraction of eligible authors who have at least one incorrect instance.

Uncertainty quantification

We use nonparametric, author-level bootstrapping with 200,000 resamples to form 95% confidence intervals. In each bootstrap replicate, authors are sampled with replacement from the relevant author set for the estimand being computed: all eligible authors for platform-specific Part 1 averages; only authors with overlapping years for the paired OpenAlex–Scopus difference; authors contributing at least one reported event for the event-level measure; and the 25-author sample (restricted to those with at least one instance) for Part 2. Within a replicate, all data from a sampled author are included, preserving within-author dependence and the OpenAlex–Scopus pairing where applicable. Estimates are recomputed on the resampled authors, and two-sided 95% percentile intervals are taken from the empirical bootstrap distribution. If a replicate contains no eligible denominator for a given estimand (e.g., no events), that replicate is redrawn for that estimand so that all intervals are based on 200,000 valid replicates.

Results

We report findings at both the institution and country levels, separately for OpenAlex (OA) and Scopus (SC). The timeline audit (Part 1) covers 100 unique authors with review data. Platform-specific denominators reflect coverage differences: 83 authors contribute OpenAlex timelines (1434 author–years), 61 contribute Scopus timelines (820 author–years), and 44 contribute to both platforms (490 overlapping author–years). Exact counts appear in Table 1, with the paired OpenAlex–Scopus comparison on overlapping years in Table 2. The publication-level audit (Part 2) covers 2093 publications across 25 authors, with 20 having at least 1 OpenAlex publication and 17 having at least one Scopus publication; mechanism-resolved error rates are in Table 3. Figures visualize point estimates with 95% bootstrap CIs.

Part 1: Timeline accuracy and migration event validity

At the *institution* level, OpenAlex shows lower accuracy than Scopus: OpenAlex author-weighted accuracy (AWA) is 0.621 (95% CI 0.546–0.693) and year-weighted accuracy (YWA) is 0.589 (0.497–0.681), whereas Scopus has AWA = 0.704 (0.627–0.776) and YWA = 0.748 (0.668–0.817). At the *country* level, accuracy is higher for both platforms but OpenAlex still trails Scopus: OpenAlex AWA = 0.750 (0.687–0.810) and YWA = 0.722 (0.633–0.803) versus Scopus AWA = 0.865 (0.814–0.911) and YWA = 0.878 (0.830–0.919). The proportion of authors with at least one incorrect year is high at institutions (OpenAlex 0.795, 0.711–0.880; Scopus 0.770, 0.656–0.869) and lower at countries (OpenAlex 0.651, 0.542–0.747; Scopus 0.508, 0.377–0.639). Full estimates and denominators are in Table 1 (OpenAlex and Scopus); Figs. 1, 2, and 3 visualize these quantities.

Restricting to overlapping in-range years within authors (44 authors; 490 author–years), the mean paired difference (OpenAlex minus Scopus) in accuracy is -0.029 for institutions (95% CI $-0.145, 0.084$) and -0.068 for countries ($-0.162, 0.014$); both intervals include zero (Table 2; Fig. 4). Event-level positive predictive value (PPV)—the fraction of reported migration events with both endpoints correct—is low for both platforms: OpenAlex PPV is 0.293 (0.214–0.382) at institutions and 0.181 (0.115–0.253) at countries; Scopus PPV is 0.451 (0.354–0.552) at institutions and 0.209 (0.110–0.317) at countries (Table 1; Fig. 5).

Part 2: Publication-level errors and mechanisms

For authors with at least one publication, OpenAlex's publication-weighted error rate (PER) for all false positives is 0.274 (95% CI 0.124–0.472) versus 1.519 (1.138–2.250) for

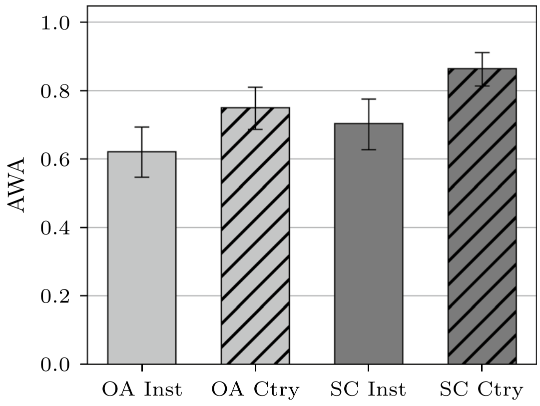

Fig. 1 Part 1: author-weighted (AWA) timeline accuracy by platform and level. *OA* OpenAlex, *SC* Scopus, *Inst* institution, *Ctry* country. Error bars: 95% bootstrap CIs ($B = 200{,}000$ author-level resamples). Numeric results are in Table 1

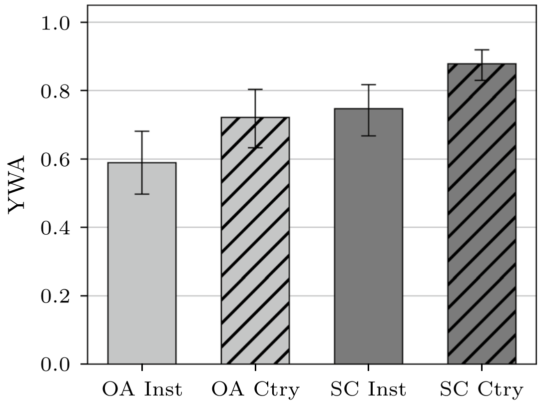

Fig. 2 Part 1: year-weighted (YWA) timeline accuracy by platform and level. Abbreviations as in Fig. 1. Numeric results are in Table 1

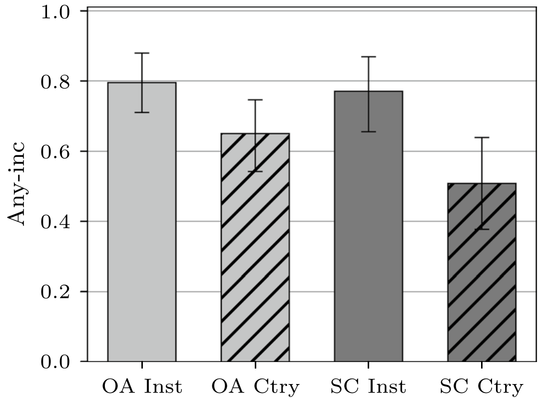

Fig. 3 Part 1: proportion of authors with any incorrect year (Any-inc) by platform and level. Abbreviations as in Fig. 1. Numeric results are in Table 1

Scopus. Decomposing mechanisms, OpenAlex's PER is 0.211 (0.047–0.428) for disambiguation errors and 0.063 (0.036–0.094) for affiliation-extraction errors, while Scopus's PER is 0.269 (0.000–1.041) and 1.250 (0.983–1.670) for these mechanisms, respectively.

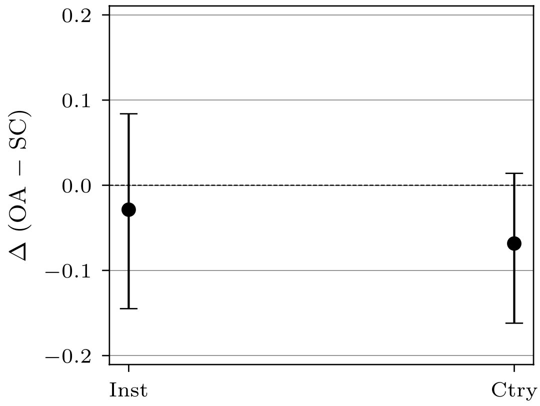

Fig. 4 Part 1: paired OA–SC accuracy difference (OA minus SC) over overlapping years, by level. Points show means; error bars show 95% bootstrap CIs. Numeric results are in Table 2

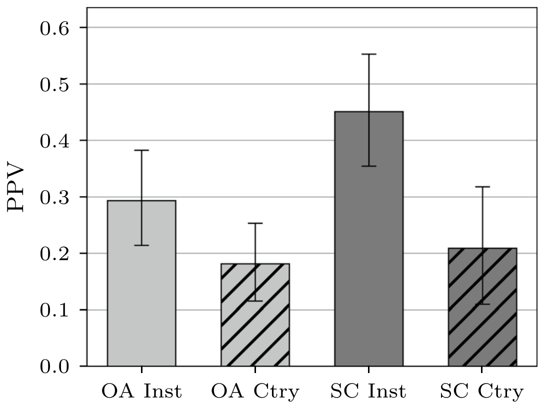

Fig. 5 Part 1: event-level positive predictive value (PPV) for migration (both endpoints correct) by platform and level. Abbreviations as in Fig. 1. Numeric results are in Table 1

Author-weighted error rates (AER) follow the same pattern (OpenAlex: 0.217 for all false positives, 0.147 disambiguation, 0.070 affiliation-extraction; Scopus: 1.565, 0.211, 1.355). The proportion of authors with any incorrect affiliation is high overall (OpenAlex 0.900, 0.750–1.000; Scopus 0.941, 0.824–1.000). All numeric results and denominators appear in Table 3. Figures 6, 7, and 8 provide visual summaries.

Discussion

This study quantified how false-positive affiliations in large bibliographic platforms propagate into researcher mobility indicators and examined the mechanisms that generate them. Using a NeurIPS 2020 cohort, we audited annual affiliation timelines for 100 authors and conducted a mechanism-resolved publication review for a focused sample. Two robust features emerge from the evidence. First, timeline accuracy is materially below one—especially at the institution level—and platform-level aggregates favor Scopus over OpenAlex. At institutions, OpenAlex's author-weighted accuracy is 0.621 and Scopus's is 0.704; at countries, OpenAlex reaches 0.750 and Scopus 0.865 (Table 1; Figs. 1, 2). The proportion of authors with at least one incorrect year follows the same hierarchy, being higher at

Table 1 Part 1: timeline accuracy and migration event validity by platform and level

Platform	Level	AWA	YWA	Any-inc	PPV	Authors	Years	Events
OA	Inst	0.621 [0.546, 0.693]	0.589 [0.497, 0.681]	0.795 [0.711, 0.880]	0.293 [0.214, 0.382]	83	1434	334
OA	Ctry	0.750 [0.687, 0.810]	0.722 [0.633, 0.803]	0.651 [0.542, 0.747]	0.181 [0.115, 0.253]	83	1434	171
SC	Inst	0.704 [0.627, 0.776]	0.748 [0.668, 0.817]	0.770 [0.656, 0.869]	0.451 [0.354, 0.552]	61	820	193
SC	Ctry	0.865 [0.814, 0.911]	0.878 [0.830, 0.919]	0.508 [0.377, 0.639]	0.209 [0.110, 0.317]	61	820	67

95% CIs in brackets

AWA author-weighted average accuracy, *YWA* year-weighted average accuracy, *Any-inc* proportion of authors with any incorrect year, *PPV* event-level positive predictive value for migration, *OA* OpenAlex, *SC* Scopus, *Inst* institution, *Ctry* country, *Authors* authors contributing to level, *Years* author–years, *Events* migration events with both endpoints in-range

Table 2 Part 1: paired OA–SC difference (OA minus SC) in accuracy over overlapping in-range years, by level

Level	Δ (OA–SC)	Authors	Overlap years
Inst	– 0.029 [– 0.145, 0.084]	44	490
Ctry	– 0.068 [– 0.162, 0.014]	44	490

95% CIs in brackets

Inst institution, *Ctry* country, *OA* OpenAlex, *SC* Scopus

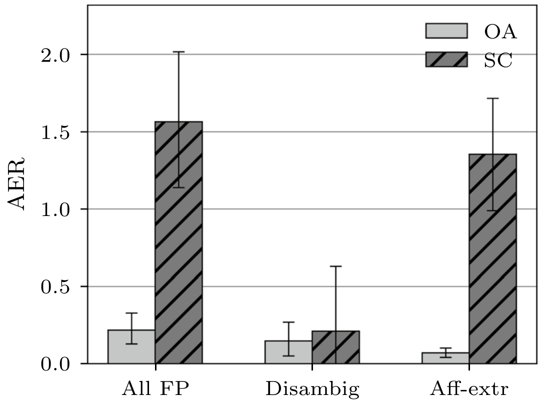

Fig. 6 Part 2: author-weighted error rates (AER) by mechanism (All FP, Disambig, Aff-extr). For each paired bar, the OpenAlex bar is on the *left* and the Scopus bar is on the *right*. Error bars: 95% bootstrap CIs. Numeric results are in Table 3

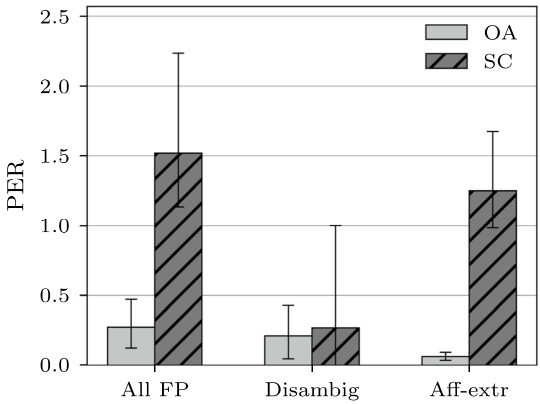

Fig. 7 Part 2: publication-weighted error rates (PER) by mechanism (may exceed 1 if multiple errors occur on the same publication). For each paired bar, the OpenAlex bar is on the *left* and the Scopus bar is on the *right*. Error bars: 95% bootstrap CIs. Numeric results are in Table 3

the institution level than at the country level for both platforms (Fig. 3). Second, reported migration events often lack fully verified endpoints. Event-level positive predictive values (PPVs) are well below one for both systems—about 0.29 for OpenAlex and 0.45 for Scopus at the institution level, and about 0.18 for OpenAlex and 0.21 for Scopus at the country level (Fig. 5; Table 1). These PPVs imply that raw event counts overstate mobility to a substantial degree.

Interpretation requires two nuances. First, the relative advantage of Scopus over OpenAlex in platform-level aggregates is attenuated when we compare platforms on common support. In the paired, within-author analysis restricted to overlapping in-range years, the

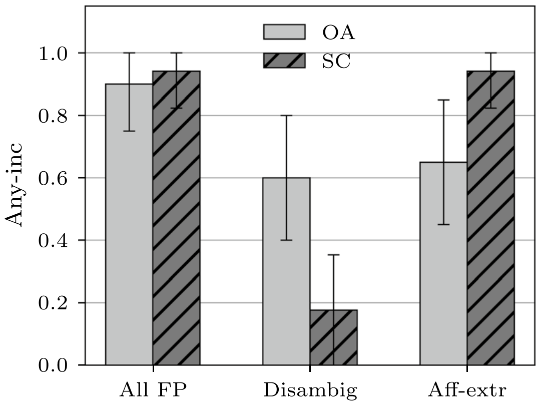

Fig. 8 Part 2: proportion of authors with any incorrect affiliation (Any-inc) by mechanism. For each paired bar, the OpenAlex bar is on the *left* and the Scopus bar is on the *right*. Error bars: 95% bootstrap CIs. Numeric results are in Table 3

Table 3 Part 2: publication-level affiliation error measures by mechanism and platform

Mechanism	Platform	AER	PER	Any-inc	Authors
All FP	OA	0.217 [0.127, 0.328]	0.274 [0.124, 0.472]	0.900 [0.750, 1.000]	20
All FP	SC	1.565 [1.143, 2.019]	1.519 [1.138, 2.250]	0.941 [0.824, 1.000]	17
Disambig	OA	0.147 [0.050, 0.268]	0.211 [0.047, 0.428]	0.600 [0.400, 0.800]	20
Disambig	SC	0.211 [0.000, 0.629]	0.269 [0.000, 1.041]	0.176 [0.000, 0.353]	17
Aff-extr	OA	0.070 [0.041, 0.101]	0.063 [0.036, 0.094]	0.650 [0.450, 0.850]	20
Aff-extr	SC	1.355 [0.989, 1.718]	1.250 [0.983, 1.670]	0.941 [0.824, 1.000]	17

95% CIs in brackets

AER author-weighted error rate, *PER* publication-weighted error rate (can exceed 1 when multiple incorrect affiliations occur on the same publication), *Any-inc* proportion of authors with any incorrect affiliation, *All FP* all false positives, *Disambig* disambiguation errors, *Aff-extr* affiliation-extraction errors, *OA* OpenAlex, *SC* Scopus

mean OpenAlex–Scopus accuracy differences center near zero and the bootstrap intervals include zero at both institution and country levels (Table 2; Fig. 4). This pattern indicates that part of the aggregate gap reflects differences in coverage and author mix rather than purely platform-specific performance. Second, PPV is higher at the institution level than at the country level for both platforms (OpenAlex: 0.293 versus 0.181; Scopus: 0.451 versus 0.209). This may seem counterintuitive given that institution-level *state* accuracy is lower. However, institution-level mode changes are more frequent—producing more migration events in the denominator—and a larger share of those events happen to have both endpoints verified. At the country level, mode changes are rarer, and when they do occur, they are more likely to involve at least one incorrect endpoint. Thus, while institution-level timelines are more fragile in terms of state accuracy, their migration events are not disproportionately spurious relative to those at the country level. Taken together, the results are consistent with the hypothesis that false positives inflate mobility indicators and with the expectation that institutional timelines are more error-prone than country-level timelines in terms of state accuracy.

The publication-level audit helps explain why these patterns arise. In OpenAlex, false positives are split between author disambiguation failures and affiliation-extraction issues (publication-weighted error rates of about 0.21 and 0.06, respectively), whereas in Scopus the dominant mechanism is affiliation extraction and normalization (about 1.25) with comparatively smaller contribution from disambiguation (about 0.27, with a wide interval that includes near-zero) (Table 3; Figs. 6–8). Values exceeding one reflect multiple incorrect affiliation instances attached to the same publication record. Mechanism matters for mobility inference: disambiguation errors fabricate entire publications for a profile, while affiliation-extraction errors mislabel otherwise valid works; both can induce spurious moves, but affiliation-extraction errors are especially consequential for institution-level timelines where small label changes can flip the annual mode.

These findings have immediate implications for measurement and reporting. Mobility analyses should present event-level validity alongside counts—PPV-weighted event summaries are a simple way to communicate the likely magnitude of overstatement without pretending to fully correct it. Where policy questions allow, country-level indicators are empirically more stable than institution-level indicators and should be preferred or, at minimum, complemented by robustness checks. Paired, within-author comparisons on overlapping years are informative about composition versus platform effects and can be incorporated operationally by requiring platform concordance on endpoints (or concordance with publication PDFs) before counting an event. Finally, the mechanism profile points to clear pipeline priorities: identity resolution remains a key target for OpenAlex, while affiliation parsing and parent–child mapping deserve emphasis for Scopus; broader use of persistent identifiers for people and organizations would benefit both.

Limitations

The empirical scope is intentionally narrow: AI researchers in a single conference year. Fields with different venue mixes or affiliation practices may exhibit different error profiles. Because NeurIPS authors are disproportionately affiliated with well-resourced institutions in North America, Europe, and East Asia and tend to have distinctive, prolific publication profiles, disambiguation may be comparatively easier in this cohort than in the broader researcher population. Consequently, the error rates reported here may understate the rates that would be observed in fields or venues with less well-documented or more ambiguous author records. The publication-level audit is designed for mechanism insight rather than population-level precision; its small size limits the sharpness of cross-platform contrasts. Ambiguous instances are excluded rather than adjudicated as errors, which likely biases accuracy upward and PPV modestly upward where uncertainty clusters around endpoints. False negatives—missing works or missing affiliations—cannot be directly observed in our framework, so reported error rates are lower bounds on the total burden of error. The exclusion of unmapped OpenAlex affiliation strings from mode computation introduces a platform asymmetry whose directional effect on accuracy estimates is ambiguous. A post hoc re-query of the OpenAlex API ("Methods" section) indicates the practical effect is limited: although 16.0% of the verified OpenAlex author-years contained at least one publication with only a raw affiliation string, just 0.7% consisted entirely of such publications and therefore had their institution mode set by carry-forward. Downstream error rates are also conditional on the accuracy of the title and author linkage step. Title matching is essentially exact, but a post hoc re-run of author matching ("Methods" section) showed that links accepted with a borderline similarity score between 0.60and 0.90are unreliable; such spurious links, although few, propagate into the affiliation review and modestly inflate the OpenAlex error rates we report. Finally, timeline reconstruction choices (mode-per-year with carry-forward and deterministic tie-breaking) are standard and pragmatic but not unique; alternative operationalizations could shift sensitivity to specific error modes.

Future work

Two extensions would strengthen generality and practice. First, replicate the audits across fields, cohorts, and years to map how mechanism prevalence shifts with coverage and curation and to test whether the institution–country accuracy gap persists. Second, integrate multi-platform concordance with persistent identifiers to raise precision at scale, and pair this with automated sequence diagnostics for career plausibility and endpoint validation. Complementary simulation studies that inject controlled disambiguation and affiliation-extraction processes into clean timelines would help calibrate how mechanism-specific error rates translate into mobility inflation under common reconstruction rules. As these tools mature, uncertainty propagation from record-level error to flow indicators should become routine in reporting.

Concluding remarks

False-positive affiliations are common, mechanism-diverse, and consequential for mobility inference. In this cohort, overall state accuracy is lower at institutions than at countries, platform-level aggregates favor Scopus over OpenAlex, and a substantial share of reported moves lack verified endpoints; yet on common support the average OpenAlex–Scopus accuracy difference recedes. The practical path forward is clear: pair event counts with validity diagnostics, prefer coarser geography where appropriate, use platform concordance to raise precision, report denominators with bootstrap uncertainty, and target improvements at the dominant mechanisms in each platform. These steps would materially improve the reliability and transparency of mobility statistics used in scientometric research and science policy.

Author contributions Mitch Mitchell led data collection and code development, contributed to methodology design, manual review, and co-wrote the manuscript. Stephanie L. Johnson led the manual review, contributed to methodology design and data interpretation, and co-wrote the manuscript. Michael D. Porter contributed to study design, supervision, writing review, and editing. All authors approved the final manuscript and agree to be accountable for all aspects of the work.

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Data availability The datasets generated and analyzed during the current study are available at https://github.com/jbmitchell02/affiliation-errors-openalex-scopus. Raw bibliographic data can be retrieved from OpenAlex and Scopus via the included scripts.

Code availability All code used for data collection and analysis is openly available at https://github.com/jbmitchell02/affiliation-errors-openalex-scopus.

Materials availability Not applicable.**Consent for publication** Not applicable.

Declarations

Conflict of interest The authors declare no conflict of interest.

Ethical approval Not applicable.

Informed consent Not applicable.

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